Practical: Extended Partition with Logical Partitions

Question

Can use any available disk (like /dev/sda, /dev/sdb) instead of /dev/nvme0n1, depending on your system's device naming.

You are given a disk /dev/sdb with free space. Perform the following:

Partitioning

1. Create **1 extended partition** on /dev/sdb
2. Inside it, create **3 logical partitions**:
   - /dev/sdb5 → **2GB**
   - /dev/sdb6 → **1GB**
   - /dev/sdb7 → **remaining space**

Formatting (different filesystems)

- Format:
  - /dev/sdb5 → **XFS**
  - /dev/sdb6 → **ext4**
  - /dev/sdb7 → **ext3**

Mounting

Temporary Mount:

- Mount /dev/sdb5 → /data1
- Mount /dev/sdb6 → /data2

Permanent Mount:

- Mount /dev/sdb7 → /backup using **UUID in /etc/fstab**

Unmount and Delete All

```
---------------commands-------------
# Verify disk
Lsblk

# Create extended + logical partitions
fdisk /dev/sdb
# inside fdisk:
# n → e → 1 → Enter → Enter
# n → Enter → +2G
# n → Enter → +1G
# n → Enter → Enter
# p → w

# Apply changes
partprobe

# Verify partitions
lsblk

# Format partitions
mkfs.xfs  /dev/sdb5
mkfs.ext4 /dev/sdb6
mkfs.ext3 /dev/sdb7

# Verify filesystem
lsblk -f

# Create mount points
mkdir /data1
mkdir /data2
```

```
mkdir /backup

# Temporary mount
mount /dev/sdb5 /data1
mount /dev/sdb6 /data2

# Verify mount
df -h
lsblk

# Get UUID
blkid

# Add permanent mount
vi /etc/fstab
# add line:
# UUID=xxxx   /backup   ext3   defaults   0 0

# Apply changes
systemctl daemon-reload
mount -a

# Final verification
df -h
lsblk -f
mount
---------------------------x-----------------------------------
```

Partitioning

```
00:56:41 | root@localhost | /mnt | cmd:129 | cpu: 0.27 | mem:43%
 | disk:21% | ip: | git: | ssh: | time:0s
🚀 lsblk
NAME         MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda            8:0    0   10G  0 disk
└─sda1         8:1    0    2G  0 part /mnt/sda1
                                      /data1
sdb            8:16   0   25G  0 disk
├─sdb1         8:17   0    1M  0 part
├─sdb2         8:18   0    2G  0 part /boot
└─sdb3         8:19   0   23G  0 part
  ├─cs-root  253:0    0 20.5G  0 lvm  /
  └─cs-swap  253:1    0  2.5G  0 lvm  [SWAP]
sr0           11:0    1  9.4G  0 rom  /run/media/root/CentOS-Stre
am-10-BaseOS-x86_64
```

3. Create **1 extended partition** on /dev/sdb

```
21:05:31 | root@localhost | ~ | cmd:144 | cpu: 0.11 | mem:27% | disk:21% |
sh: | time:0s
🚀 fdisk /dev/sda

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

This disk is currently in use - repartitioning is probably a bad idea.
It's recommended to umount all file systems, and swapoff all swap
partitions on this disk.


Command (m for help): p

Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010

Device     Boot Start     End Sectors Size Id Type
/dev/sda1        2048 4196351 4194304   2G 83 Linux
```

```
Command (m for help): n
Partition type
   p   primary (1 primary, 0 extended, 3 free)
   e   extended (container for logical partitions)
Select (default p): e
Partition number (2-4, default 2): 2
First sector (4196352-20971519, default 4196352):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (4196352-20971519, default 20971519): +5G

Created a new partition 2 of type 'Extended' and of size 5 GiB.

Command (m for help): p
Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010

Device     Boot   Start      End  Sectors Size Id Type
/dev/sda1          2048  4196351  4194304   2G 83 Linux
/dev/sda2       4196352 14682111 10485760   5G  5 Extended
```

4. Inside it, create **3 logical partitions**:

   - /dev/sdb5 → **2GB**

```
Command (m for help): n
Partition type
   p   primary (1 primary, 1 extended, 2 free)
   l   logical (numbered from 5)
Select (default p): l

Adding logical partition 5
First sector (4198400-14682111, default 4198400):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (4198400-14682111, default 14682111): +2G

Created a new partition 5 of type 'Linux' and of size 2 GiB.

Command (m for help): p
Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010

Device     Boot   Start      End  Sectors Size Id Type
/dev/sda1          2048  4196351  4194304   2G 83 Linux
/dev/sda2       4196352 14682111 10485760   5G  5 Extended
/dev/sda5       4198400  8392703  4194304   2G 83 Linux
```

   - /dev/sdb6 → **1GB**

```
Command (m for help): n
Partition type
   p   primary (1 primary, 1 extended, 2 free)
   l   logical (numbered from 5)
Select (default p): l

Adding logical partition 6
First sector (8394752-14682111, default 8394752):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (8394752-14682111, default 14682111): +1G

Created a new partition 6 of type 'Linux' and of size 1 GiB.

Command (m for help): p
Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010

Device     Boot   Start      End  Sectors Size Id Type
/dev/sda1          2048  4196351  4194304   2G 83 Linux
/dev/sda2       4196352 14682111 10485760   5G  5 Extended
/dev/sda5       4198400  8392703  4194304   2G 83 Linux
/dev/sda6       8394752 10491903  2097152   1G 83 Linux
```

- /dev/sdb7 → **remaining space**

```
Command (m for help): n
Partition type
   p   primary (1 primary, 1 extended, 2 free)
   l   logical (numbered from 5)
Select (default p): l

Adding logical partition 7
First sector (10493952-14682111, default 10493952):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (10493952-14682111, default 14682111): +3G
Value out of range.
Last sector, +/-sectors or +/-size{K,M,G,T,P} (10493952-14682111, default 14682111):

Created a new partition 7 of type 'Linux' and of size 2 GiB.

Command (m for help): p
Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010

Device     Boot    Start      End  Sectors Size Id Type
/dev/sda1           2048  4196351  4194304   2G 83 Linux
/dev/sda2        4196352 14682111 10485760   5G  5 Extended
/dev/sda5        4198400  8392703  4194304   2G 83 Linux
/dev/sda6        8394752 10491903  2097152   1G 83 Linux
/dev/sda7       10493952 14682111  4188160   2G 83 Linux
```

```
Command (m for help): w
The partition table has been altered.
Syncing disks.
```

```
21:09:02 | root@localhost | ~ | cmd:144 | cpu: 0.08 | mem:27% |
 | git: | ssh: | time:0s
🚀 partprobe
```

Formatting (different filesystems)

```
21:14:56 | root@localhost | ~ | cmd:147 | cpu: 0.08 | mem:27% | disk:21% | ip:
 | git: | ssh: | time:0s
🚀 mkfs.
mkfs.cramfs  mkfs.ext2    mkfs.fat     mkfs.vfat
mkfs.erofs   mkfs.ext3    mkfs.minix   mkfs.xfs
mkfs.exfat   mkfs.ext4    mkfs.msdos
```

- Format:
    - /dev/sdb5 → **XFS**

```
21:09:12 | root@localhost | ~ | cmd:145 | cpu: 0.15 | mem:27% | disk:21% | ip:
 | git: | ssh: | time:0s
🚀 mkfs.xfs /dev/sda5
meta-data=/dev/sda5              isize=512    agcount=4, agsize=131072 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=1
         =                       reflink=1    bigtime=1 inobtcount=1 nrext64=1
         =                       exchange=0   metadir=0
data     =                       bsize=4096   blocks=524288, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1, parent=0
log      =internal log           bsize=4096   blocks=16384, version=2
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
         =                       rgcount=0    rgsize=0 extents
         =                       zoned=0      start=0 reserved=0
```

    - /dev/sdb6 → **ext4**

```
21:14:34 | root@localhost | ~ | cmd:146 | cpu: 0.00 | mem:26% | disk:21% | i
 | git: | ssh: | time:0s
🚀 mkfs.ext4 /dev/sda6
mke2fs 1.47.1 (20-May-2024)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: 7a022a13-b8bb-4c95-b35a-8d6ba22a17ee
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
```

    - /dev/sdb7 → **ext3**

```
21:14:56 | root@localhost | ~ | cmd:147 | cpu: 0.08 | mem:27% | disk:21% | ip:
 | git: | ssh: | time:0s
🚀 mkfs.ext3 /dev/sda7
mke2fs 1.47.1 (20-May-2024)
Creating filesystem with 523520 4k blocks and 131072 inodes
Filesystem UUID: baa6e4a8-bfb8-487a-b3bd-621f0cafc3d0
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
```

```
 | git: | ssh: | time:0s
🚀 lsblk -f
NAME FSTYPE FSVER LABEL UUID
MOUNTPOINTS
sda
├─sda1
│     xfs                  a1b98820-f4c5-43ad-ab2c-b5585b83d06f
/mnt/sda1
├─sda2
│
├─sda5
│     xfs                  25d620b1-8ad9-4d1d-a365-20ed23d72477
├─sda6
│     ext4   1.0           7a022a13-b8bb-4c95-b35a-8d6ba22a17ee
└─sda7
      ext3   1.0           baa6e4a8-bfb8-487a-b3bd-621f0cafc3d0
```

Mounting

```
21:16:08 | root@localhost | ~ |
 | git: | ssh: | time:0s
🚀 mkdir /{data1,data2}
```

Temporary Mount:

- Mount /dev/sdb5 → /data1
- Mount /dev/sdb6 → /data2

```
21:18:47 | root@localhost | ~ | cmd:151 |
 | git: | ssh: | time:0s
🚀 mount /dev/sda5 /data1

21:19:16 | root@localhost | ~ | cmd:152 |
 | git: | ssh: | time:0s
🚀 mount /dev/sda6 /data2
```

Permanent Mount:

```
21:23:27 | root@localhost | ~ | cmd:155 | cpu: 0.22
 | git: | ssh: | time:0s
🚀 mkdir /backup
```

- Mount /dev/sdb7 → /backup using **UUID in /etc/fstab**

```
21:23:46 | root@localhost | ~ | cmd:156 | cpu: 0.62 | mem:27% | disk:21% | ip:
 | git: | ssh: | time:0s
🚀 blkid
/dev/mapper/cs-root: UUID="e79aa976-2f04-4e51-8942-08170334e649" BLOCK_SIZE="5
12" TYPE="xfs"
/dev/sdb3: UUID="JvsA7G-OIq1-xO2O-iLGb-yedU-K6uc-VWRD31" TYPE="LVM2_member" PA
RTUUID="2b5c3d62-67af-4337-a7b7-7f8d918ea142"
/dev/mapper/cs-swap: UUID="de55cc71-342c-4646-817d-ae6a99e96439" TYPE="swap"
/dev/sdb2: UUID="b41c8b19-dfe8-4088-8806-ec8fa03b629a" BLOCK_SIZE="512" TYPE="
xfs" PARTUUID="16b9aa22-0b3a-4829-b097-3d46938ee14b"
/dev/sdb1: PARTUUID="443939da-3edc-452b-a7c8-0c04c6dd7d17"
/dev/sda2: PTTYPE="dos" PARTUUID="17c80010-02"
/dev/sda7: UUID="baa6e4a8-bfb8-487a-b3bd-621f0cafc3d0" SEC_TYPE="ext2" BLOCK_S
IZE="4096" TYPE="ext3" PARTUUID="17c80010-07"
/dev/sda5: UUID="25d620b1-8ad9-4d1d-a365-20ed23d72477" BLOCK_SIZE="512" TYPE="
xfs" PARTUUID="17c80010-05"
/dev/sda1: UUID="a1b98820-f4c5-43ad-ab2c-b5585b83d06f" BLOCK_SIZE="512" TYPE="
xfs" PARTUUID="17c80010-01"
/dev/sda6: UUID="7a022a13-b8bb-4c95-b35a-8d6ba22a17ee" BLOCK_SIZE="4096" TYPE=
"ext4" PARTUUID="17c80010-06"
```

```
# /etc/fstab
# Created by anaconda on Sun Mar 15 19:41:01 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=e79aa976-2f04-4e51-8942-08170334e649 /                       xfs     defaults        0 0
UUID=b41c8b19-dfe8-4088-8806-ec8fa03b629a /boot                   xfs     defaults        0 0
UUID=de55cc71-342c-4646-817d-ae6a99e96439 none                    swap    defaults        0 0


/dev/sda1       /mnt/sda1       xfs     defaults        0       0
#-------------------------------------------------------------------------------

#/dev/sda7: UUID="baa6e4a8-bfb8-487a-b3bd-621f0cafc3d0" SEC_TYPE="ext2" BLOCK_SIZE="4096" TYPE="ext3" PARTUUID="17c80010-07"

#/dev/sda7      /backup ext3    defaults        0       0
UUID=baa6e4a8-bfb8-487a-b3bd-621f0cafc3d0      /backup ext3    defaults        0       0
```

```
21:25:50 | root@localhost | ~ | cmd:158 | cpu: 0.11 | mem:27% | disk:21% |
p: | git: | ssh: | time:0s
🚀 systemctl daemon-reload

21:26:12 | root@localhost | ~ | cmd:159 | cpu: 0.08 | mem:27% | disk:21% |
p: | git: | ssh: | time:0s
🚀 mount -a
```

```
21:26:59 | root@localhost | ~ | cmd:161 | cpu: 0.22 | mem:27% | disk:21% | i
p: | git: | ssh: | time:0s
🚀 df -h
Filesystem           Size  Used Avail Use% Mounted on
/dev/mapper/cs-root   21G  4.3G   17G  21% /
devtmpfs             2.8G     0  2.8G   0% /dev
tmpfs                2.8G   84K  2.8G   1% /dev/shm
tmpfs                1.2G  9.5M  1.2G   1% /run
tmpfs                1.0M     0  1.0M   0% /run/credentials/systemd-journald
.service
/dev/sdb2            2.0G  446M  1.6G  23% /boot
tmpfs                570M  116K  570M   1% /run/user/0
/dev/sda1            2.0G   71M  1.9G   4% /mnt/sda1
/dev/sda5            2.0G   71M  1.9G   4% /data1
/dev/sda6            974M   24K  907M   1% /data2
/dev/sda7            2.0G   60K  1.9G   1% /backup
```

```
21:28:03 | root@localhost | ~ | cmd:162 | cpu: 0.15 | mem:27% | disk:21% | ip: | git: | ssh: | time:0s
🚀 lsblk -f
NAME FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda
├─sda1
│    xfs                a1b98820-f4c5-43ad-ab2c-b5585b83d06f    1.9G     4% /mnt/sda1
├─sda2
│
├─sda5
│    xfs                25d620b1-8ad9-4d1d-a365-20ed23d72477    1.9G     4% /data1
├─sda6
│    ext4   1.0         7a022a13-b8bb-4c95-b35a-8d6ba22a17ee  906.2M     0% /data2
└─sda7
     ext3   1.0         baa6e4a8-bfb8-487a-b3bd-621f0cafc3d0    1.8G     0% /backup
```

Unmount and Delete All

```
21:29:31 | root@localhost | ~ | cmd:164 | cpu: 0.10 | mem:27% |
 | git: | ssh: | time:0s
🚀 umount /dev/sda7

21:45:18 | root@localhost | ~ | cmd:165 | cpu: 0.01 | mem:27% |
 | git: | ssh: | time:0s
🚀 umount /dev/sda6

21:45:23 | root@localhost | ~ | cmd:166 | cpu: 0.01 | mem:27% |
 | git: | ssh: | time:0s
🚀 umount /dev/sda5

21:45:33 | root@localhost | ~ | cmd:167 | cpu: 0.01 | mem:27% |
 | git: | ssh: | time:0s
🚀 vim /etc/fstab
```

```
#
# /etc/fstab
# Created by anaconda on Sun Mar 15 19:41:01 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=e79aa976-2f04-4e51-8942-08170334e649 /                       xfs     defa
ults        0 0
UUID=b41c8b19-dfe8-4088-8806-ec8fa03b629a /boot                   xfs     defa
ults        0 0
UUID=de55cc71-342c-4646-817d-ae6a99e96439 none                    swap    defa
ults        0 0


/dev/sda1       /mnt/sda1       xfs     defaults        0       0
#------------------------------------------------------------------------------
---

~
~
~
~
```

```
21:46:08 | root@localhost | ~ | cmd:168 | cpu: 0.00 | mem:27% | disk:21% |
 | git: | ssh: | time:0s
🚀 systemctl daemon-reload
```

```
21:46:39 | root@localhost | ~ | cmd:170 | cpu: 0.00 | mem:27% | disk:21% |
 | git: | ssh: | time:0s
🚀 fdisk /dev/sda

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

This disk is currently in use - repartitioning is probably a bad idea.
It's recommended to umount all file systems, and swapoff all swap
partitions on this disk.


Command (m for help): p

Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010

Device     Boot    Start      End  Sectors Size Id Type
/dev/sda1           2048  4196351  4194304   2G 83 Linux
/dev/sda2        4196352 14682111 10485760   5G  5 Extended
/dev/sda5        4198400  8392703  4194304   2G 83 Linux
/dev/sda6        8394752 10491903  2097152   1G 83 Linux
/dev/sda7       10493952 14682111  4188160   2G 83 Linux
```

```
Command (m for help): d
Partition number (1,2,5-7, default 7): 7

Partition 7 has been deleted.

Command (m for help): d
Partition number (1,2,5,6, default 6): 6

Partition 6 has been deleted.

Command (m for help): d
Partition number (1,2,5, default 5): 5

Partition 5 has been deleted.

Command (m for help): d
Partition number (1,2, default 2): 2

Partition 2 has been deleted.

Command (m for help): p
Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010

Device     Boot Start     End Sectors Size Id Type
/dev/sda1        2048 4196351 4194304   2G 83 Linux

Command (m for help): w
The partition table has been altered.
```

```
21:48:01 | root@localhost | ~ | cmd:171 | cpu: 0.00 | mem:27% | disk:21% |
 | git: | ssh: | time:0s
🚀 partprobe
```

```
21:48:07 | root@localhost | ~ | cmd:172 | cpu: 0.00 | mem:28% | disk:21% | ip
 | git: | ssh: | time:0s
🚀 lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda           8:0    0   10G  0 disk
└─sda1        8:1    0    2G  0 part /mnt/sda1
sdb           8:16   0   25G  0 disk
├─sdb1        8:17   0    1M  0 part
├─sdb2        8:18   0    2G  0 part /boot
└─sdb3        8:19   0   23G  0 part
  ├─cs-root 253:0    0 20.5G  0 lvm  /
  └─cs-swap 253:1    0  2.5G  0 lvm  [SWAP]
sr0          11:0    1 1024M  0 rom

21:50:35 | root@localhost | ~ | cmd:173 | cpu: 0.05 | mem:27% | disk:21% | ip
 | git: | ssh: | time:0s
🚀 df -h
Filesystem           Size  Used Avail Use% Mounted on
/dev/mapper/cs-root   21G  4.3G   17G  21% /
devtmpfs             2.8G     0  2.8G   0% /dev
tmpfs                2.8G   84K  2.8G   1% /dev/shm
tmpfs                1.2G  9.5M  1.2G   1% /run
tmpfs                1.0M     0  1.0M   0% /run/credentials/systemd-journald.s
ervice
/dev/sdb2            2.0G  446M  1.6G  23% /boot
tmpfs                570M  116K  570M   1% /run/user/0
/dev/sda1            2.0G   71M  1.9G   4% /mnt/sda1
```